

# SOURCE-PRIVATE TOOL-TRACE PACKETS FOR RATE-CAPPED EVIDENCE COMMUNICATION

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## ABSTRACT

Agent systems can benefit when one model uses private evidence observed by another without relaying a full trace. We formalize this as source-private evidence communication with decoder side information: a target agent receives the public task and a candidate pool, while a source agent observes private execution evidence and sends a rate-capped message. We instantiate this setting as an explicit diagnostic-code protocol benchmark where the target selects among public repair candidates and the source observes private tool-trace diagnostics. A compact explicit REPAIR\_DIAG packet, decoded against candidate-side decoder information, improves target-side candidate selection from 25% to 80.8–100.0% across four frozen 500-example core and held-out-family surfaces. Zero-source, shuffled-source, random same-byte, answer-only, answer-masked, target-derived, same-byte structured relay, helper/no-log, and trace-removed controls remain near target-only (25.0–25.8% in the reported rows). Model-produced packets average 1.55–2.00 bytes, compared with roughly 366–374 bytes and 34 tokens for full hidden-log relay. A small Qwen3 model-mediated protocol-decoder sanity check improves over target/control on 16–32 examples, but remains a smoke ablation. The result is a narrow but reproducible positive method: explicit source-private tool-trace packets can transmit the benchmark’s hidden diagnostic evidence under the controls tested here and with explicit rate accounting.

## 1 INTRODUCTION

Multi-agent and cross-model systems increasingly divide work across components with asymmetric information. A tool-using source agent may observe execution traces, retrieval results, private memory, or environment feedback unavailable to the target agent that must make the final decision. Full trace relay is a simple interface, but it can be wasteful when the target already has useful side information such as a candidate pool or cached prior.

We study a narrower question: can a source agent send a compact, auditable message that changes the target’s decision only when matched private source evidence is present? This framing follows distributed source coding and rate-distortion intuitions: a sender can communicate fewer bits when the decoder has side information, and the objective is task distortion at a communication rate rather than reconstruction of the whole source state (Slepian & Wolf, 1973; Wyner & Ziv, 1976; Shannon, 1959). Here, the target’s candidate pool is decoder side information, and the source packet is evaluated by downstream selector accuracy under source controls.

We instantiate this idea in hidden-repair candidate selection. The target receives a public repair issue, buggy implementation, and four public repair candidates. The source receives a private hidden-test/tool trace containing an explicit repair diagnostic. The source model emits a compact REPAIR\_DIAG packet, and the target-side decoder uses the packet and public candidate decoder side information to select a repair candidate. The packet is interpretable and byte-counted.

The contribution is intentionally scoped. We do not claim unstructured latent transfer, raw-log repair reasoning, or universal cross-model communication. We claim that a rate-capped source-private packet can carry the benchmark’s hidden diagnostic evidence to a target-side candidate decoder under the controls tested here. The contribution is a falsifiable communication protocol benchmark, not evidence of emergent latent alignment.

Source-private evidence communication with decoder side information

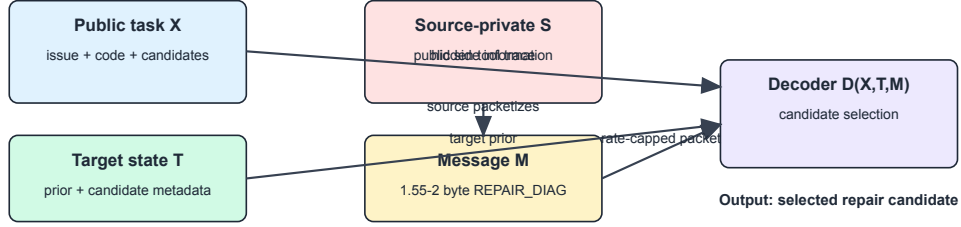


Figure 1: Source-private evidence communication with decoder side information. The source observes private evidence  $S$  and sends a rate-capped packet  $M$ . The target combines public state  $X$ , candidate state  $T$ , and  $M$  to select a candidate.

This makes the method closer to a syndrome-like evidence packet than to prompt compression or hidden-state transfer: the packet is only useful because the target already has candidate-side decoder side information that can decode it. The novelty is not discovering an unknown semantic representation; it is a rate-capped, source-private evidence interface with falsification controls.

## 2 PROBLEM SETUP

Each example has public task and candidate information  $X$ , target-side state  $T$ , source-private evidence  $S$ , a rate-capped source message  $M$ , and a decoder  $D(X, T, M)$  that selects a final candidate. The target never sees  $S$  directly except in full-relay oracle baselines. A method is counted as communication only if  $D(X, T, M_{\text{matched}})$  improves over no-source and source-destroying controls: zero-source, shuffled-source, random same-byte, answer-only, answer-masked, target-derived, and same-byte relay messages.

We report selector accuracy, candidate-pool recall, exact ID parity, packet validity, paired uncertainty, bytes/tokens, and source-control deltas.

## 3 BENCHMARK

The hidden-repair benchmark contains deterministic Python repair tasks, in the spirit of test-guided program-repair workflows where tests help localize and validate defects (Hu et al., 2025). Each example has public issue text, a buggy implementation, four public candidate repairs, one gold candidate in the pool, and a source-private hidden-test/tool log. Public candidate decoder side information includes `handles.repair_diag=<code>`. The private source log contains `private.tool_trace: REPAIR_DIAG=<code>`.

By construction, the target prior solves exactly one quarter of examples. The main surfaces are a strict-small 160-example slice, a 500-example core slice, a 500-example held-out repair-family slice, and seed repeats over core seeds 29/31 and held-out seeds 30/32. Deterministic surfaces have candidate-pool recall 1.0, so reported improvements are selector improvements.

The synthetic design is deliberate: it gives deterministic hidden evidence, exact ID parity, complete source-destroying controls, and byte-level accounting. The constructional 25% target floor isolates source-private evidence transfer from candidate generation. It is not intended to prove unconstrained program repair or real deployment generality.

| Surface     | Source model | Matched | Target  | Control | Valid | Bytes | $\Delta$ 95% CI |
|-------------|--------------|---------|---------|---------|-------|-------|-----------------|
| core 29     | Qwen3-0.6B   | 404/500 | 125/500 | 126/500 | 0.776 | 1.55  | [0.516,0.600]   |
| core 29     | Phi-3-mini   | 500/500 | 125/500 | 126/500 | 1.000 | 2.00  | [0.714,0.788]   |
| core 31     | Qwen3-0.6B   | 404/500 | 125/500 | 128/500 | 0.776 | 1.55  | [0.516,0.602]   |
| core 31     | Phi-3-mini   | 500/500 | 125/500 | 128/500 | 1.000 | 2.00  | [0.710,0.786]   |
| held-out 30 | Qwen3-0.6B   | 461/500 | 125/500 | 129/500 | 0.864 | 1.73  | [0.632,0.712]   |
| held-out 30 | Phi-3-mini   | 500/500 | 125/500 | 129/500 | 1.000 | 2.00  | [0.710,0.788]   |
| held-out 32 | Qwen3-0.6B   | 462/500 | 125/500 | 126/500 | 0.860 | 1.72  | [0.634,0.716]   |
| held-out 32 | Phi-3-mini   | 500/500 | 125/500 | 126/500 | 1.000 | 2.00  | [0.710,0.786]   |

Table 1: Model-mediated source-packet rows over four frozen 500-example surfaces. Control is the highest-accuracy source-destroying control among the reported control set. Intervals are paired bootstrap 95% intervals for matched minus target.

| Surface     | Matched | Target | Zero  | Shuffled | Random | JSON  | Free text | Diag-masked |
|-------------|---------|--------|-------|----------|--------|-------|-----------|-------------|
| core 29     | 1.000   | 0.250  | 0.250 | 0.250    | 0.254  | 0.250 | 0.250     | 0.250       |
| held-out 30 | 1.000   | 0.250  | 0.250 | 0.250    | 0.254  | 0.250 | 0.250     | 0.250       |

Table 2: Protocol/evaluator threat-model rows at the 2-byte packet budget. Same-byte JSON/free-text relays and diagnostic-masked logs stay at target-only.

## 4 METHOD

The source model receives a private hidden-test log and emits only a compact repair packet. In the main model-mediated condition, `trace_no_hint`, the private diagnostic line remains in the private log, but the copied helper line and hint are removed. Invalid packets fall back to the target prior.

The target-side protocol decoder receives the public candidate pool and packet. If the packet is a valid diagnostic matching exactly one candidate’s `handles_repair_diag` decoder-side-information metadata, the decoder selects that candidate; otherwise it returns the target-prior candidate. This design makes the communication channel auditable, but it is protocol-shaped rather than a learned latent bridge. We therefore include a Qwen3 target-decoder smoke ablation in Section 9.

## 5 BASELINES AND CONTROLS

We compare against target-only and target-wrapper/no-source baselines; zero, shuffled, random same-byte, and target-derived source-destroying controls; answer-only and answer-masked leakage controls; matched-byte hidden-log prefix, JSON, and free-text relays; full hidden-log and full diagnostic oracles; and trace-component ablations including diagnostic-masked, expected/actual-masked, test-name-masked, and trace-removed logs.

## 6 RESULTS

Table 1 reports model-produced source packets across four frozen 500-example surfaces. Qwen3 reaches 80.8–92.4% accuracy and Phi-3 reaches 100%, while target-only remains 25% and source-destroying controls remain near target-only.

Removing the private trace diagnostic kills the effect: Qwen3 `raw_log_no_trace` returns to 25% with zero valid packets on all four surfaces.

## 7 THREAT MODEL RESULTS

At the 2-byte packet budget, source-destroying and leakage controls remain at target-only while full-log and diagnostic oracles reach 100%. Table 2 summarizes representative core and held-out surfaces.

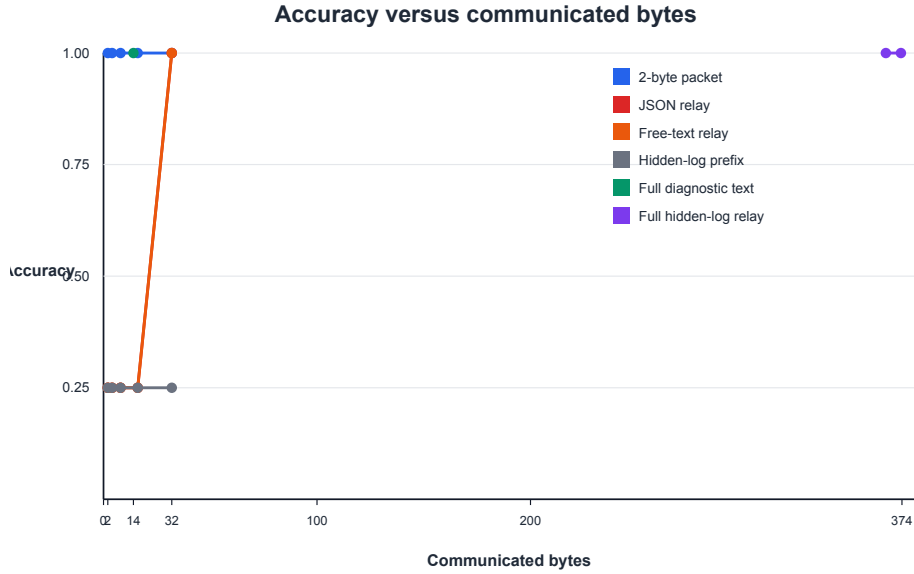


Figure 2: Accuracy versus communicated bytes, averaged over representative core and held-out reviewer-risk surfaces. This is a low-rate advantage, not dominance at all budgets: structured text relay becomes an oracle when granted enough bytes, while compact packets occupy the far-left-rate regime.

## 8 RATE AND SYSTEMS ANALYSIS

The method’s main systems-relevant property is the far-left-rate operating point. Model-produced packets average 1.55–2.00 bytes and roughly one token under the tokenizer/accounting used in these runs. Full diagnostic text is also an oracle but costs roughly 14 bytes; full hidden-log relay reaches oracle accuracy at roughly 366–374 bytes and 34 tokens per example. Structured JSON and free-text relays fail at 2 bytes but become oracles at 32 bytes because the diagnostic is then visible in parseable form.

## 9 MODEL-MEDIATED PROTOCOL DECODER SMOKE

As a limited check on whether the result depends entirely on the hard-coded target lookup, we replace it with Qwen3-0.6B acting as a target-side selector. The model receives the target-prior fallback label, source packet, candidate labels, and diagnostic decoder side information. On a 16-example core slice, matched packets reach 0.688 versus 0.250 target/control. On a 32-example held-out slice, matched packets reach 0.750 versus 0.250 target and 0.281 best control. This is a smoke ablation, not the main evidence or a learned latent bridge, but it is consistent with the protocol being usable by an LLM selector.

## 10 INTERPRETABILITY

The packet channel is directly auditable in this benchmark. The packet is a two-character diagnostic code; target candidate decoder side information states which diagnostic each candidate handles; and component-masking controls identify the private trace field that carries the signal. Failures can be audited as invalid source packets, fallback-to-prior cases, or target-decoder selection mistakes.

## 11 RELATED WORK

**Source coding and side information.** Our setup is closest to source coding with decoder side information: Slepian-Wolf and Wyner-Ziv show that communication can be reduced when the decoder has correlated side information (Slepian & Wolf, 1973; Wyner & Ziv, 1976), and rate-distortion theory frames performance as task distortion at a communication rate (Shannon, 1959). Recent distributed indirect source-coding work is also relevant because the sender observes evidence about a task latent rather than transmitting a full reconstructable state (Tang et al., 2024). Semantic communication motivates task-oriented utility per transmitted bit (Xie et al., 2021).

**Tool and agent communication.** Tool-using agents expose observations and execution traces that may be private to one component of a system (Yao et al., 2023; Schick et al., 2023). Multi-agent frameworks motivate explicit handoff contracts between specialized agents (Wu et al., 2024). Our benchmark turns this handoff into a controlled source-private channel: the receiver sees the public task and candidate pool, while only the source sees the hidden tool-trace diagnostic.

**Prompt compression and text relay.** Prompt-compression methods such as LLMingua reduce visible context length (Jiang et al., 2023). They are important budget baselines, but they address a different setting from ours when the target never receives the hidden evidence. We therefore compare against matched-byte text, JSON, hidden-log prefix, full diagnostic text, and full hidden-log relay rows to separate byte savings from private evidence communication.

**Latent, cache, and connector communication.** Recent preprints on direct activation and KV-cache communication, such as C2C and KVComm, communicate internal model state rather than explicit packets (Fu et al., 2026; Shi et al., 2026). VLM connectors and JEPA-style objectives motivate future learned bottlenecks and anti-collapse diagnostics (Alayrac et al., 2022; Li et al., 2023; Assran et al., 2023; Bardes et al., 2022). The present method is deliberately narrower: it sends an auditable source-private diagnostic packet and evaluates whether matched source evidence, rather than target priors or extra prompt bytes, changes the final selection.

## 12 LIMITATIONS

The main limitation is scope. Candidate decoder side information exposes the diagnostic mapping, and the primary decoder is deterministic and protocol-shaped. The target-decoder smoke reduces this concern but remains small. The benchmark is synthetic and candidate-selection based; it does not establish unconstrained program repair, raw-log reasoning, real tool/retrieval deployment, or learned latent transfer. Structured text relay becomes an oracle when granted enough bytes. Stronger future comparisons should include learned prompt compression, C2C/KVComm-style cache transfer, and larger model-mediated target decoders under the same source-private controls.

## 13 CONCLUSION

We present a scoped positive method for source-private evidence communication. A source model reads private tool-trace diagnostics and emits a compact explicit packet. A target-side candidate decoder combines that packet with public candidate decoder side information to select the gold candidate from the provided repair pool. The reported result is artifact-backed, interpretable, rate-accounted, and stable across the tested frozen 500-example surfaces, held-out repair-family slices, seed repeats, source-destroying controls, and structured relay baselines. A small target-LLM decoder smoke ablation provides a limited sanity check.

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| Candidate | Public decoder side information | Patch intent                   |
|-----------|---------------------------------|--------------------------------|
| patch 0   | handles X0                      | strict key access              |
| patch 1   | handles H1                      | missing-key default            |
| patch 2   | handles X2                      | numeric fallback; target prior |
| patch 3   | handles X3                      | uppercase status               |

Table 3: Representative source-private example. Without matched private evidence the target prior selects patch 2; with packet H1, the decoder selects the gold candidate, patch 1.

| Evidence               | N / seeds                               | Entry point                                      | Artifact         |
|------------------------|-----------------------------------------|--------------------------------------------------|------------------|
| Model packet rows      | 500; 29/31/30/32                        | model packet extractor and baseline-pack builder | baseline-pack.md |
| Reviewer controls      | 500; 29 and 30                          | hidden-repair packet evaluator                   | manifest.md      |
| Target-decoder smoke   | core 16 seed 29; held-out 16/32 seed 30 | target-decoder smoke runner                      | manifest.md      |
| Figures and rate curve | 500; 29 and 30                          | figure builder                                   | manifest.json    |
| LaTeX compile          | paper source                            | latexmk                                          | manifest.md      |

Table 4: Decisive artifact manifests. Manifests and paired JSON/Markdown summaries record paths, seeds, budgets, summaries, and where available exact commands; the git commit provides source hashes.

## A CLAIM BOUNDARY

We claim that explicit source-private tool-trace packets can communicate hidden diagnostic evidence to a target-side candidate decoder under rate constraints. We do not claim raw-log repair inference, learned latent transfer, universal cross-model communication, or unconstrained repair generation.

## B REPRESENTATIVE EXAMPLE

Example sphr\_0001 asks for reading key status from a mapping and returning unknown when the key is absent. The target sees only the public issue and candidate pool; its prior selects a numeric-fallback patch. The source-private hidden test observes an empty input, expected value unknown, and a KeyError. The private trace line contains REPAIR\_DIAG=H1, so the matched packet is H1. The protocol decoder then selects the candidate whose public decoder side information handles H1.

## C ARTIFACTS

Main result artifacts are tracked in the baseline-pack, reviewer-risk, target-decoder-smoke, and LaTeX/figure result roots, all prefixed by results/source\_private\_tool\_trace\_\*. Table 4 lists the decisive manifests and scripts.

The reproduction entry points are the hidden-repair packet generator/evaluator, model packet extractor, baseline-pack builder, figure builder, and target-decoder smoke scripts; exact commands and paths are recorded in the artifact manifests.